

Time allowed: 85 minutes.

All questions are to be attempted

Each question is to be handed in separately.

Section A (START A NEW PAGE)

Question 1 :

(a) If $z = -2 + 2i$, express in the form $x + iy$.

(i) $\bar{z}$

(ii) $\frac{1}{z}$

(iii) z^2

(b) Plot on the Argand diagram the points z , $\bar{z}$, $\frac{1}{z}$, clearly showing the relations between their modulus and arguments.

Question 2 :

A parallelogram ABCD is drawn with $\vec{OA} = 3 + 2i$, $\vec{OC} = 11 + 9i$ and $\vec{AB} = 2 + 5i$.
Find the vectors $\vec{OB}$ and $\vec{OD}$.

Question 3 :

(i) Write in mod-arg form $4\sqrt{3} - 4i$

(ii) hence describe the transformation z_1 to z_2 if $z_2 = (4\sqrt{3} - 4i) z_1$.

(iii) If $z_1 = 2 + 2\sqrt{3}i$, find the point z_2 .

Section B (START A NEW PAGE)

Question 1 :

(i) Describe the locus $|z - 2 + i| = 3$.

(ii) Hence find the maximum value of $|z|$.

Question 2 :

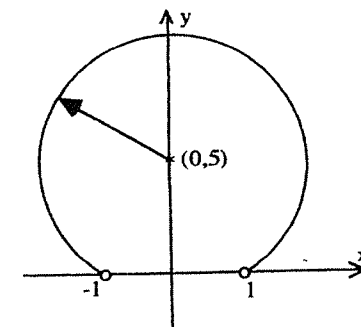
(i) Graph the locus $\text{Arg}(z - 3 + i) = -\frac{\pi}{4}$.

(ii) Graph the locus $\text{Im}(z^2) = 8$.

Question 3 :

The locus of a point z is shown. (see diagram)

Evaluate correct to 2 decimal places:
 $\text{Arg}(z - 1) - \text{Arg}(z + 1)$.



Question 4 :

(i) Find the Cartesian equation for the locus $|z - 1 + 2i| = 3|z + 3 - 4i|$.

(ii) Describe the locus.

Section C : (START A NEW PAGE)

Question 1:

(i) Find the exact square roots of $16 - 30i$, and

(ii) Hence solve: $z^2 - (3 - i)z - 2 + 6i = 0$.

Question 2 :

(i) Solve $z^4 + 16 = 0$.

(ii) Plot the zeros of $z^4 + 16$ on a circle on the Argand diagram if

$$\text{Arg}(z_1) < \text{Arg}(z_2) < \text{Arg}(z_3) < \text{Arg}(z_4) < \pi.$$

(iii) Show graphically the vector addition of $z_1 + z_2 + z_3 + z_4$.

(iv) Evaluate $\text{Arg}(z_1 - z_3) - \text{Arg}(z_2 - z_4)$.

(v) Factorise $z^4 + 16$ over $\mathbb{R}$.

Section D (START A NEW PAGE)

Question 1 :

(i) Given that z is a complex number with $|z| = 1$, use DeMoivre's Theorem to prove that $\cos n\theta = \frac{z^n + z^{-n}}{2}$.

(ii) Hence deduce: $\cos\theta \cos 2\theta = \frac{1}{2}(\cos 3\theta + \cos\theta)$

Question 2 :

(i) Prove $|z|^2 = z\bar{z}$

(ii) A parallelogram OABC has diagonals intersecting at M.

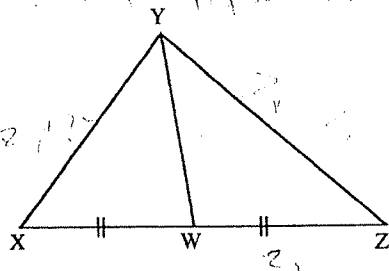
Let OA be z_1 , OC be z_2 and $0 < \text{Arg}(z_1) < \text{Arg}(z_2) < \frac{\pi}{2}$.

Draw the parallelogram OABC clearly showing the point M and the vectors z_1 , z_2 , $z_1 + z_2$, $z_1 - z_2$.

(iii) Prove that: $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2|z_1|^2 + 2|z_2|^2$.

(iv) Hence prove Apollonius' Theorem for a triangle XYZ

$$XY^2 + YZ^2 = 2(YW^2 + XW^2)$$



THIS IS THE END OF THE PAPER

$$z^n = 1 \cos n\theta$$

$$z^{-n} = 1 \cos(-n\theta)$$

$$z^n + z^{-n} = (\cos n\theta + \cos(-n\theta))^{1/2}$$

$$= \cos \frac{2n\theta}{2} + i \sin \frac{2n\theta}{2}$$

$$= \cos n\theta + i \sin n\theta$$

$$= e^{in\theta}$$

$$2 \cos n\theta = z^n + z^{-n}$$

$$\cos n\theta = \frac{z^n + z^{-n}}{2}$$

$$2 \cos \theta \cos 2\theta = \cos 3\theta + \cos \theta$$

$$\cos \theta \cos 2\theta = \frac{1}{2}(\cos 3\theta + \cos \theta)$$

13 (i) $\bar{z} = -2 - 2i$

(ii) $\frac{1}{z} = \frac{1}{-2+2i} \times \frac{2+2i}{2+2i}$

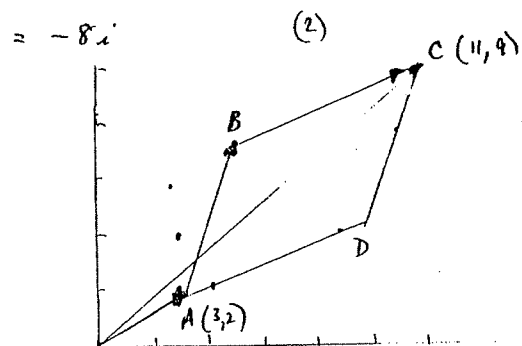
$$= \frac{2+2i}{8}$$

$$= \frac{1}{4} + \frac{1}{4}i$$

(iii) $z^2 = (-2+2i)^2$

$$= 4 - 8i - 4$$

$$= -8i$$



$$\vec{OB} = \vec{OA} + \vec{AB}$$

$$= 3+2i + 2+5i$$

$$= 5+7i$$

$$\vec{OD} = \vec{OC} - \vec{CD}$$

$$= 11+4i - (2+5i)$$

$$= 9-1i$$

13 (i) $4\sqrt{3} - 4i$

$|z| = 8$ $\text{Arg } z = -\frac{\pi}{6}$

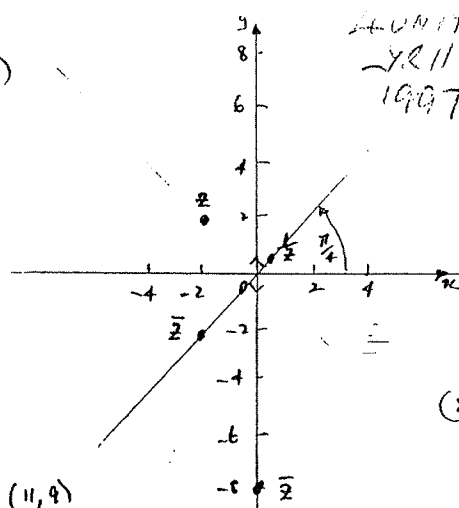
$$z = 8 \left[\cos\left(-\frac{\pi}{6}\right) + j \sin\left(-\frac{\pi}{6}\right) \right]$$

(ii) $z_1 \rightarrow z_2$ Modulus of z_1 is increased by 8 and its argument decreased by $\frac{\pi}{6}$.

(iii) $z_1 = (4\sqrt{3} - 4i)(2 + 2\sqrt{3}i)$

$$= 8\sqrt{3} + 24i - 8i + 8\sqrt{3}$$

$$= 16\sqrt{3} + 16i$$

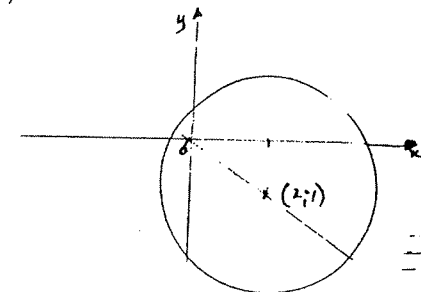


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circle centre $(2, -1)$ radius 3 units

(ii) $\text{Max } |z| = \sqrt{2^2 + (-1)^2} + 3$

$$= 3 + \sqrt{5} \text{ units.}$$



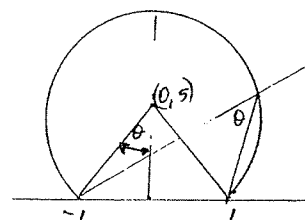
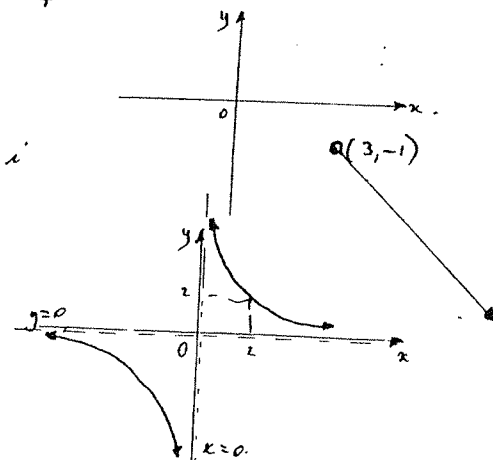
(i) $\text{Arg}(z - 3 + i) = -\frac{\pi}{4}$

(ii) $z^2 = x^2 - y^2 + 2xyi$

$\therefore \text{Im}(z^2) = 8$

$$2xy = 8$$

$$xy = 4$$



$\text{Arg}(z-1) - \text{Arg}(z+1) = \theta$ (shown)

$$\tan \theta = \frac{1}{5}$$

$\theta = 0.120$ (2 dec places)

4

$$\frac{|z-1+2i|}{|z+3-4i|} = 3$$

$$(x-1)^2 + (y+2)^2 = 9[(x+3)^2 + (y-4)^2]$$

$$x^2 - 2x + 1 + y^2 + 4y + 4 = 9[x^2 + 6x + 9 + y^2 - 8y + 16]$$

$$8x^2 + 8y^2 + 56x - 76y + 220 = 0$$

$$x^2 + 7x + \frac{49}{4} + y^2 - \frac{19y}{2} + \frac{361}{4} = \frac{55}{2} + \frac{49}{4} + \frac{361}{4}$$

$$\left(x + \frac{7}{2}\right)^2 + \left(y - \frac{19}{4}\right)^2 = \frac{117}{16}$$

Locus: circle
centre $\left(-\frac{7}{2}, \frac{19}{4}\right)$
radius $\frac{3\sqrt{13}}{4}$

$$(x+iy)^2 = 16-30i$$

$$x^2 - y^2 = 16$$

$$2xy = -30$$

$$y = -\frac{15}{x}$$

$$x^2 - \frac{225}{x^2} = 16$$

$$x^4 - 16x^2 - 225 = 0$$

$$(x^2 - 25)(x^2 + 9) = 0$$

$$x = \pm 5 \text{ only } x \in \mathbb{R}. \quad x^2 + 9 = 0 \text{ No real soln.}$$

$$x = 5 \quad y = -3$$

$$x = -5 \quad y = 3$$

$$\therefore \text{roots are } \pm (5-3i)$$

$$\text{ii) } z^2 - (3-i)z - 2+6i = 0$$

$$z = \frac{3-i \pm \sqrt{(3-i)^2 - 4(-2+6i)}}{2}$$

$$= \frac{3-i \pm \sqrt{9-6i-1+8-24i}}{2}$$

$$= \frac{3-i \pm \sqrt{16-30i}}{2}$$

$$= \frac{3-i \pm (5-3i)}{2}$$

$$= \frac{8-4i}{2} \quad \frac{-2+2i}{2}$$

$$z = 4-2i, -1+i$$

(2)

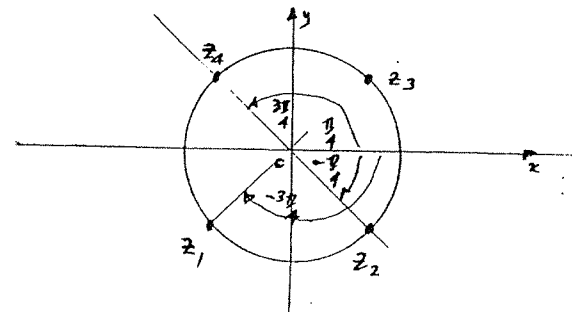
$$z^4 = -16$$

$$z^4 = 16 \operatorname{cis}(\pi + 2\pi n) \quad n \text{ integer.}$$

$$\therefore z = 2 \operatorname{cis}\left(\frac{\pi + 2\pi n}{4}\right)$$

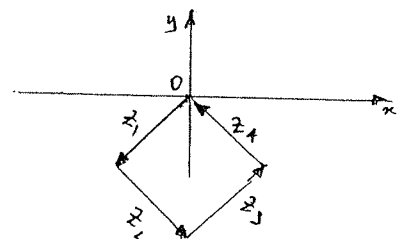
$$z = 2 \operatorname{cis} \frac{\pi}{4}, 2 \operatorname{cis} \frac{3\pi}{4}, 2 \operatorname{cis} -\frac{\pi}{4}, 2 \operatorname{cis} -\frac{3\pi}{4}$$

(ii)



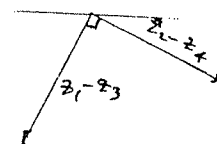
(2)

(iii)



(2)

(iv)



$$\operatorname{Arg}(z_1 - z_3) - \operatorname{Arg}(z_2 - z_4) = -\frac{\pi}{2}$$

$$\begin{aligned} \text{iv) } z^4 + 16 &= \left(z - 2\left[\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right]\right) \left(z - 2\left[\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right]\right) \left(z - 2\left[\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4}\right]\right) \left(z - 2\left[\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4}\right]\right) \\ &= \left[(z - 2\cos \frac{\pi}{4})^2 + 4\sin^2 \frac{\pi}{4}\right] \left[(z - 2\cos \frac{3\pi}{4})^2 + 4\sin^2 \frac{3\pi}{4}\right] \\ &= \left[z^2 - 4\cos \frac{\pi}{4}z + 4\cos^2 \frac{\pi}{4} + 4\sin^2 \frac{\pi}{4}\right] \left[z^2 - 4\cos \frac{3\pi}{4}z + 4\cos^2 \frac{3\pi}{4} + 4\sin^2 \frac{3\pi}{4}\right] \\ &= \left[z^2 - 4 \cdot \frac{1}{\sqrt{2}}z + 4\right] \left[z^2 - 4 \cdot \left(-\frac{1}{\sqrt{2}}\right)z + 4\right] \\ &= [z^2 - 2\sqrt{2}z + 4][z^2 + 2\sqrt{2}z + 4] \end{aligned}$$

$$\begin{aligned} \text{or } z^4 + 16 &= (z^2 + 4)^2 - 8z^2 \\ &= (z^2 + 4 + 2\sqrt{2}z)(z^2 + 4 - 2\sqrt{2}z) \end{aligned}$$

(2)

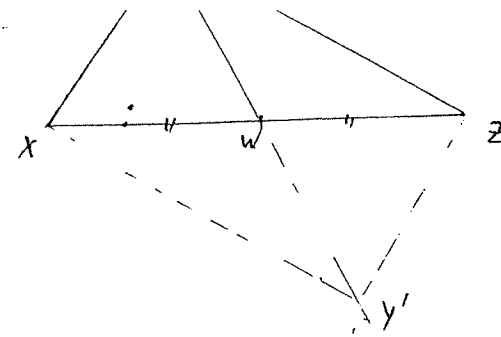
$$(\cos \theta + i \sin \theta)^{-n} = \cos n\theta - i \sin n\theta$$

$$\therefore \text{If } z = \cos \theta + i \sin \theta \quad |z| = 1$$

$$z^n + z^{-n} = 2 \cos n\theta$$

$$\cos n\theta = \frac{z^n + z^{-n}}{2}$$

(2)



Form parallelogram
XYZY'

$$\text{From (iii)} \quad XZ^2 + (YY')^2 = 2XY^2 + 2YZ^2$$

$$(2XW)^2 + (2YW)^2 = 2XY^2 + 2YZ^2$$

$$4XW^2 + 4YW^2 = 2XY^2 + 2YZ^2$$

$$XY^2 + YZ^2 = 2XW^2 + 2YW^2$$

$$XY^2 + YZ^2 = 2[XW^2 + YW^2]$$

(3)

$$\text{ii) } \cos \theta \cos 2\theta = \frac{1}{2}(z + z^{-1}) \cdot \frac{1}{2}(z^2 + z^{-2})$$

$$= \frac{1}{4} [z^3 + z^{-1} + z + z^{-3}]$$

$$= \frac{1}{2} \left[\frac{z^3 + z^{-3}}{2} + \frac{z + z^{-1}}{2} \right]$$

(3)

$$\text{iv) } \cos \theta \cos 2\theta = \frac{1}{2}(\cos 3\theta + \cos \theta)$$

$$\text{(i) If } z = x + iy \text{ then } \bar{z} = x - iy$$

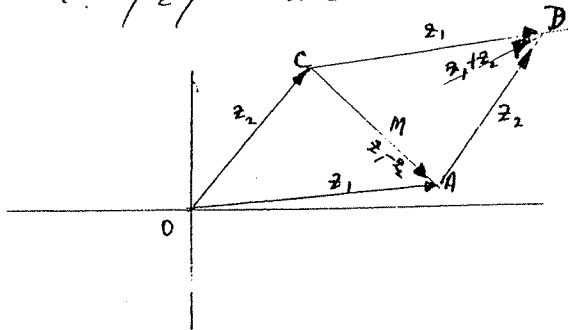
$$z\bar{z} = (x + iy)(x - iy)$$

$$= x^2 + y^2$$

$$= |z|^2$$

(2)

$$\therefore |z|^2 = z\bar{z}$$



(2)

$$\begin{aligned} \text{ii) } |z_1 + z_2|^2 + |z_1 - z_2|^2 &= (z_1 + z_2)(\bar{z}_1 + \bar{z}_2) + (z_1 - z_2)(\bar{z}_1 - \bar{z}_2) \\ &= z_1\bar{z}_1 + z_1\bar{z}_2 + z_2\bar{z}_1 + z_2\bar{z}_2 + z_1\bar{z}_1 - z_1\bar{z}_2 - z_2\bar{z}_1 + z_2\bar{z}_2 \\ &= |z_1|^2 + |z_2|^2 + |z_1|^2 + |z_2|^2 \\ &= 2|z_1|^2 + 2|z_2|^2 \end{aligned}$$

(3)